

Mya Bolds, Ph.D. Student

Computer Science and Engineering Department
CSE Department, University of California, San Diego, La Jolla, CA
mbolds@ucsd.edu | www.myabolds.me

Education

University of California San Diego Doctorate of Computer Science	<i>La Jolla, CA</i> Expected 2028
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University of California San Diego Master of Science in Computer Science	<i>La Jolla, CA</i> 2025
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Xavier University of Louisiana Bachelor of Computer Science	<i>New Orleans, LA</i> 2021
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Skills

Programming & Tools: Python, Java, SQL, R, PySpark, Databricks, Unity Catalog, LangChain, Swift, React Native, Optuna, Azure ML, MLflow, TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, Dlib, MATLAB, Model Context Protocol, Git, Matplotlib, Seaborn, Plotly, Pandas
Machine Learning: Computer Vision, Natural Language Processing, Vision-Language Models, Transfer Learning, Fine-tuning, Prompt Engineering, Hyperparameter Tuning, Agentic Modeling, MLOps

Research & Analysis: Experimental Design, Quantitative Analysis, Qualitative Research, Data Visualization, Data Normalization, Data Pre-processing, Automated Pipeline Engineering, Web Scraping, Cross-functional Collaboration, Production Readiness Assessment, Technical Communication

Awards and Fellowships

Gem Fellow	Present
UC-HBCU Fellow	Present
Sloan Scholar	Present
McNair Scholar	2018-2021
Norman C. Francis Full-Tuition Scholarship	2017-2021

Professional Experience

Xerox-Lexmark <i>Data Scientist Intern</i>	Lexington, KY June 2025- September 2025
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- Designed and implemented an agentic automation prototype in Databricks using PySpark, SQL, Python, and LangChain to streamline data cleaning in the Medallion Architecture.
- Fine-tuned Microsoft's Florence VLM on Azure ML to classify and diagnose print quality errors, improved inference quality through prompt engineering, and applied MLOps practices using MLflow for experiment tracking and Optuna/manual hyperparameter optimization.
- Researched agentic automation via Claude and MCP servers, designing prompt-driven pipelines for automating data cleaning and ingestion tasks.
- Participated in cross-functional team meetings to assess model pipeline production readiness.

University of California-San Diego

Graduate Student Researcher

La Jolla, CA

August 2022- Present

- Designing a domain adaptation framework to mitigate bias in AI
- Led a mixed-methods study that involves training nine paired conditional GAN models to replicate Instagram beauty filter transformations and leading a student team in developing a Smart Mirror system (Java, HTML, CSS frontend; Python/Flask backend; SQL + Supabase database) to deploy models in real time, conduct user surveys, and audit how filter-induced changes shift AI predictions of race, age, gender, and emotion.
- Evaluating AI agents for bias and developing agents to evaluate agentic bias.
- Designed a bias auditing pipeline that uses Computer Vision, python, pretrained models, and statistical modeling to quantify Instagram beauty filter effects on facial features before and after filter application, while examining the influence of Western beauty standards in shaping these modifications.

University of California-San Diego

Teaching Assistant – Deep Learning

La Jolla, CA

March 2026-June 2026

- Held weekly office hours and graded assignments covering core deep learning architectures (multi-layer perceptrons, CNNs, RNNs) and foundational methods (backpropagation, automatic differentiation).
- Designed homework assignments on Recurrent Neural Networks and delivered lectures on Convolutional Neural Network architecture to support undergraduate instruction in deep learning fundamentals.

University of California-San Diego

Teaching Assistant – Machine Learning for Music

La Jolla, CA

March 2025-June 2025

- Designed and maintained course notebooks that applied core ML and statistical methods (logistic regression, SVMs, k-NN, TF-IDF, stratified k-fold cross-validation) to support instruction.
- Created technical slide content on machine learning fundamentals(linear regression, neural networks, activation functions, convolutional neural networks, and evaluation metrics).

University of California-San Diego

Teaching Assistant – Interaction Design Research(HCI)

La Jolla, CA

September 2024-December 2024

- Facilitated student learning in interaction design topics, including prototyping, social computing, and user interface evaluation.
- Reviewed and provided feedback on student-led discussions and paper commentaries to improve critical analysis and engagement.

Xerox-Lexmark

Data Engineer Intern

Remote

June 2022- August 2022

- Transformed raw data into a customer-preferred format and tasked with creating a data pipeline to cleanse data via Databricks' Medallion architecture methodology using Python and SQL.
- Assisted with moving data from the Big Decisions platform to the Databricks platform.

Xavier University of Louisiana Computer Science Department

Build Research Technician

Hybrid

June 2021- June 2022

- Used the Flickr faces dataset to observe the difference in facial recognition accuracy between lighter and darker skin tones.
- Noted false positives within the Haar Cascades facial recognition algorithm.
- Attempted to detect textured/coily hair in images using edge detection and by comparing images of afro hair to straight hair using the Structural Similarity Index.

University of California San Diego STARS Research Program

Research Intern

La Jolla, CA

June 2019- August 2019

- Assisted in the development of a framework to locate the position of a drone via an attached microchip with precision.
- Wrote a python script to identify the location of a microchip.
- Used Kalman Filter to fuse information from an Inertial Measurement Unit and microchip to obtain an improved estimation of the drone's position.

Published Work

Naba Rizvi, William Wu, Mya Bolds, Andrew Begel, Imani Munyaka, and Raunak Mondal, "Are Robots Ready to Deliver Autism Inclusion?" *ACM Special Interest Group on Computer-Human Interaction(SIGCHI)*, 2024.

Naba Rizvi, Taggert Smith, Tanvi Vidyala, Mya Bolds, Andrew Begel, Imani Munyaka, Harper Strickland, and Rua Williams, "“I Hadn't Thought About That”: Creators of Human-like AI Weigh in on Ethics & Neurodivergence," *ACM Conference on Fairness, Accountability, and Transparency(FACCT)*, 2025.

Mya Bolds, Kristen Vaccaro, and, Imani Munyaka " What Race Labels Miss: Continuous Appearance Measures for Auditing Computer Vision Systems," *In submission*, 2025.

Mya Bolds, Kristen Vaccaro, and, Imani Munyaka " The Mirror Lies: How Beauty Filters Rewrite Race, Age, and Gender for AI Systems," *In submission*, 2026.

Imani N. S. Munyaka, Alexis Morales Flores, Mya Bolds ” Centering Context in Computing Ethics Education: An Experience Report,” *Conference on Research in Equity and Sustained Participation in Engineering, Computing, and Technology(RESPECT)*, 2026

Relevant Coursework

Advanced Computer Vision
Theoretical Aspects of Large Language Models
Computer Vision II
Web Mining & Recommender Systems(Machine Learning)
Statistical Natural Language Processing
Computer Vision I
Principles of Software Development
Interaction Design Research(HCI)

Organizations

National Society of Black Engineers , Member	Joined 2019
IEEE , Member	Joined 2024
AnitaB.org , Member	Joined 2024

Service

Girls Inc Workshop	La Jolla, CA
<i>Educational Workshop Leader</i>	July 2024

- Facilitated a Capture the Flag (CTF) workshop, designed to teach high school girls from underrepresented backgrounds about cyber-safety through interactive challenges.

FANCY Girls Workshop	San Diego, CA
<i>Digital Safety Instructor</i>	April 2024

- Organized a small-scale Capture the Flag (CTF) workshop, designed to teach middle school girls from underrepresented backgrounds about online safety through hands-on activities.

University of California-San Diego STARS Mentor Program	La Jolla, CA
<i>Graduate Advocate Mentor</i>	June 2022- August 2022

- Mentored undergraduate HBCU students and offered guidance on Ph.D. program preparation.
- Conducted weekly meetings where I presented information about addressing imposter syndrome, presenting at conferences, crafting research abstracts, reading and writing research papers, and enhancing social media presence.